Assignment 1 - Researching an agentic framework application

Submission Date: Sunday, September 13th, 2026

Think about how you can apply an Agentic GeoAI framework to your own academic or professional research interests.

Create a 6-minute presentation where you introduce your background, formulate a meaningful geographic research problem, and propose a small agentic application that could be developed further in this class, for example, as the basis of your final coursework project.

The emphasis is on thoughtful problem formulation and responsible system design, not on building a finished application. Your proposal should make clear which decisions belong to the human researcher, which tasks could reasonably be delegated to an agent, and how you would determine whether the resulting analysis is correct and defensible.

Table 1 outlines the important framework dimensions to consider.

Framework component	Question to address
Objective and stopping condition	What specific outcome should the agent produce, and how will it know the task is finished?
User and decision context	Who will use the output, and what decision or research activity will it support?
Inputs and provenance	Which spatial and non-spatial data are required? Where do they come from, and what quality or licensing issues matter?
Agent workflow	Show the sequence of planning, acting, observing results, validating, and revising or stopping.
Tools	Which GIS operations, code libraries, databases, APIs, models, or other tools may the agent use?
Human–agent responsibilities	Which bounded tasks may be delegated, and which scientific, ethical, or interpretive decisions must remain with a person?
Outputs and claim boundary	What artifacts will be created, and what conclusions may be drawn from them?
Validation	What automated checks, independent calculations, visual inspections, comparison data, or expert review would provide evidence that the result is correct?
Failure modes and safeguards	How could the workflow fail through bad data, hallucinated information, spatial error, bias, privacy issues, unsafe actions, or misleading communication? How will these risks be limited?
Scope and cost	How will you keep the prototype feasible, reproducible, and economical in its use of model tokens, web searches, tools, and data?

Table 1 Framework aspects to consider.

Please submit your presentation online as a pdf. You will then present your pitch in the next class.

Table 2 then outlines the assessment criteria to consider.

Finally, students may use AI tools in accordance with the course AI policy. Yet, students remain responsible for understanding, testing, validating and documenting all submitted material. The paper must include a brief AI-use statement identifying the tools used and explaining how they contributed. Undisclosed AI-generated material, fabricated citations and other materials, or code that the student cannot explain may constitute an Honor Code violation and will be reported.

Criterion	Evidence	Weight
Personal and course connection	The presentation clearly links the student's background and learning goals to the proposed work.	15%
Research problem and significance	The topic is geographically meaningful and identifies relevant users or stakeholders.	25%
Agentic framework	The proposed workflow clearly defines inputs, tools, agent actions, human responsibilities, outputs, and a stopping condition.	30%
Responsible design and validation	The proposal identifies limitations and failure modes and offers credible checks, safeguards, and claim boundaries.	20%
Communication	Slides are clear and legible, sources are acknowledged, and the presentation stays within the assigned time and scope.	10%

Table 2 Assessment criteria.